

SPENCER FOLK, Ph.D.spencer.folk@gmail.comspencerfolk.github.io**EDUCATION**

2019-2026	University of Pennsylvania <i>Ph.D., Mechanical Engineering (Advisors: Dr. Vijay Kumar, Dr. Mark Yim)</i> <i>M.S. Robotics</i>	Philadelphia, PA
2015-2019	Lafayette College <i>B.S., Mechanical Engineering (Summa Cum Laude)</i>	Easton, PA

SKILLS

Perception, Learning, & Control	Receding Horizon Optimal Control (MPC), System Identification, Nonlinear Filtering, Sensor Fusion, Deep Reinforcement Learning, Vision-based Navigation & Control
AI/ML Paradigms	Deep Learning (Transformers, GNNs, CNNs, MLP), RAG, PyTorch
Programming & Deployment	Python, C++, MATLAB, ROS 2, GPU-Accelerated Deployment (NVIDIA Platforms), Real-Time Embedded Systems, PX4, Docker, Git
Simulation, Design, & Analysis	Gazebo, RotorPy (Creator/Maintainer), SOLIDWORKS, OpenFOAM

RESEARCH EXPERIENCE

2026-present	AI/ML Applied Researcher <i>Autonomy Division, Aurora Flight Sciences</i>	Cambridge, MA
	- Engineered a monocular vision-based relative pose estimation system built on hierarchical localization to eliminate AprilTags for the DARPA LINC program, tested on an NVIDIA Jetson (invention disclosure pending). - Built an internal agentic AI workflow tool using custom web scrapers and text embedding to enable personalized RAG-based LLM querying and analysis of government solicitations. - Served as technical lead on proposal efforts for DARPA and USSOCOM JATF, architecting the core technical approaches. - Implemented a thermal estimation algorithm in PX4 for fixed-wing UAVs, validated in both software- and processor-in-the-loop (SIL/PIL) test environments. - Trained graph neural networks (GNNs) in simulation to predict the roles and organizational structure of drone swarms based purely on kinematic observations.	
2022-2025	Pathways Intern – Research Aerospace Engineer <i>Ames Research Center, National Aeronautics and Space Administration (NASA)</i>	Mountain View, CA
	- Developed and deployed a GPU-accelerated receding horizon optimal controller on flight-ready hardware, leveraging local wind predictions from neural networks to reduce UAV energy consumption and crash rates by up to 30% in windy urban environments. - Devised a deep learning-based framework for predicting wind flow fields in real time on computationally constrained aerial vehicles using on-board range and wind measurements. - Architected the end-to-end software stack for autonomous flight testing, executing the first hardware experiments in a novel open-air wind tunnel facility. - Implemented lean computational fluid dynamics code to rapidly generate large urban wind datasets for deep neural network training.	

- 2019-2025** **Graduate Research Assistant – Aerial Robotics** Philadelphia, PA
GRASP Lab, University of Pennsylvania
- Developed energy-efficient, wind-aware motion planners to safely navigate multirotor UAVs in real time through cluttered, wind-disturbed urban environments.
 - Created [RotorPy](#), an open-source aerial robotics simulator used in over 40 research papers to date and adopted for teaching topics in motion planning, state estimation, and control.
 - Collaborated on continuous trajectory-aware controller adaptation frameworks using meta-learning, reducing tracking errors despite unmodeled and/or changing dynamics.
 - Tooled nonlinear filtering methods for online system identification of UAV aerodynamics and actuator dynamics, enabling adaptive control schemes and fault detection.
- 2019** **Undergraduate Research on Ethorobotics** Easton, PA
Department of Mechanical Engineering, Lafayette College
- Modeled the movement of Archerfish in schooling and hunting sequences with Markov Decision Processes and stochastic differential equations.
 - Replicated fish swimming on a 5-DoF robotic fish to successfully elicit specific social behaviors among Archerfish, culminating in a journal paper in *Biological Cybernetics*.
- 2017-2019** **Design and Fabrication of 3D-Printed UAVs** Aberdeen, MD
Vehicle Technology Directorate, United States Army Research Laboratory (ARL)
- Designed, manufactured, and assembled size-scalable 3D-printed quadrotor UAVs that reduced part and fastener count by 58%, satisfying new Army requirements for a mission-specific UAV available on demand within 24 hours of request.
 - Developed a routine using textbook rotorcraft mechanics that appropriately sized and outfitted a quadrotor to optimize for mission parameters like range, endurance, loiter time, and payload capacity.
 - Applied deep learning to identify max endurance and max range flight speeds for 3D-printed UAVs based on flight data, enabling efficient flight for aircraft accounting for often unmodeled effects like profile drag, rotor-rotor interference, and rotor-frame interference.

SELECT PUBLICATIONS

Folk, S., Melton, J., Margolis, B. W. L., Yim, M., & Kumar, V. (2025). Towards Safe and Energy-Efficient Real-Time Motion Planning in Windy Urban Environments. *International Conference on Robotics and Automation (ICRA) 2025*. [Conference]

Folk, S., Melton, J., Margolis, B. W. L., Yim, M., & Kumar, V. (2024). Learning Local Urban Wind Flow Fields from Range Sensing. *IEEE Robotics and Automation Letters*. [Journal]

Sanghvi, H., **Folk, S.,** & Taylor, C. J. (2024). OCCAM: Online Continuous Controller Adaptation with Meta-Learned Models. *Conference on Robot Learning (CoRL)*. [Conference]

Weakly, J., Li, X., Agarwal, T., Li, M., **Folk, S.,** Jiang, C., & Sung, C. (2024). Bistable Aerial Transformer: A Quadrotor Fixed-Wing Hybrid That Morphs Dynamically Via Passive Soft Mechanism. *Journal of Mechanisms and Robotics*, 16(7), 071016. [Journal]

Folk, S., Paulos, J., & Kumar, V. (2023). RotorPy: A Python-based Multirotor Simulator with Aerodynamics for Education and Research. *The Role of Robotic Simulators for Unmanned Aerial Vehicles – ICRA 2023*. [Conference Workshop]

Brown, A. A., Brown, M. F., **Folk, S. R.,** & Utter, B. A. (2021). Archerfish respond to a hunting robotic conspecific. *Biological Cybernetics*, 115(6), 585-598. [Journal]

INVITED TALKS

“Towards Safe and Efficient Real-Time Motion Planning in Windy Urban Environments”

2025 INFORMS Annual Meeting

October 2025

East Coast Meetup, Dronecode Foundation, Philadelphia, PA.

March 2025

MEAM Departmental Seminar Series, University of Pennsylvania, Philadelphia, PA.

February 2025

Nikolai Matni Group, University of Pennsylvania, Philadelphia, PA.

October 2024

“Learning-Based Methods for Real-Time Wind Prediction in Urban Environments”

Intelligent Robot Motion Lab x FAST Group, Princeton University, Princeton, NJ.

May 2024

LEADERSHIP AND MENTORING

2020-2025 Virtual High School Mentoring

Polygence

- Fostered curiosity and interest in STEM among high school students by guiding them through self-motivated projects, which have included designing a gimbaled rocket, solar-powered UAV, and an aerial landmine detection system.

2020-2022 Teaching Assistantships

University of Pennsylvania

- Graduate courses: Advanced Robotics (600 level).
- Undergraduate courses: Control for Autonomous Robots (400 level), Vibrations (300 level), Statics and Strengths of Materials (200 level).

2021 Vice President, Mechanical Engineering Graduate Association

University of Pennsylvania

- Upheld a sense of community through the COVID-19 pandemic by organizing virtual events ranging from Zoom trivia nights to wellness challenges.

2018-2019 Chapter President, Tau Beta Pi (PA-E)

Lafayette College

- Organized chapter and community events, communicated with headquarters to maintain national standards, and represented the chapter at national and district conferences.

HONORS AND AWARDS

2019 Graduate Assistance in Areas of National Need (GAANN) Fellowship

University of Pennsylvania

Awarded to “graduate students of superior ability planning to pursue the highest degree in their course of study.”

2019 Karl J. Ammerman Prize

Lafayette College

Awarded to the “most deserving student in the mechanical engineering department, as selected by the faculty of the department.”
